

# Figures for the paper

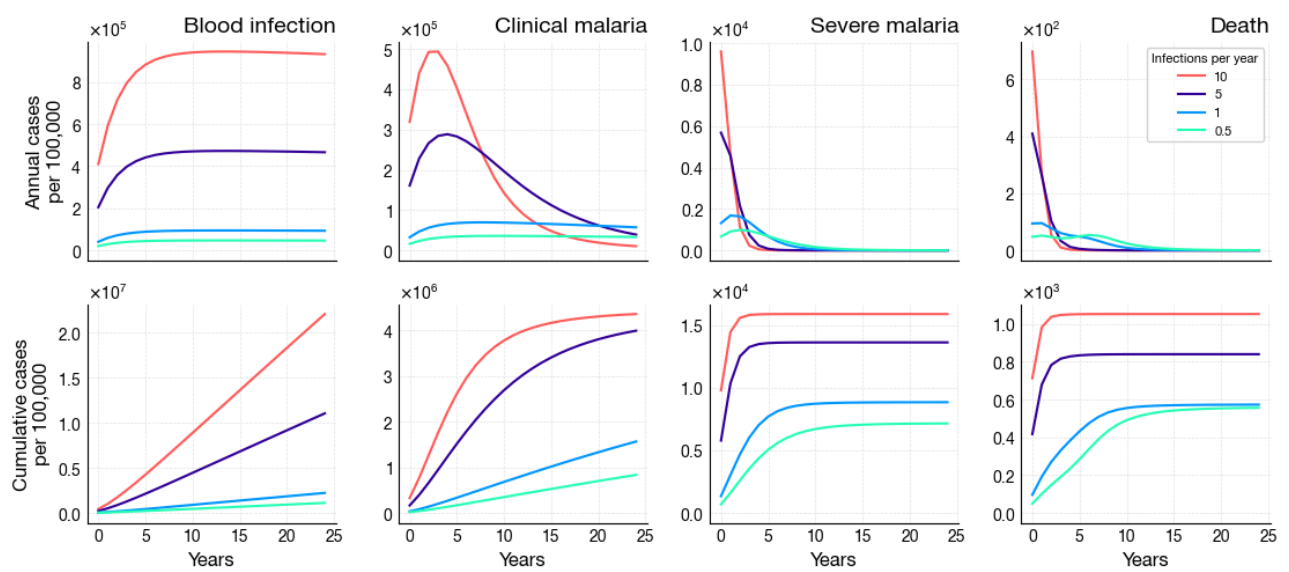
Nicholas J. Savill

Philip J. Spence

| Figure           | Description                                                                                                                                   |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Supplementary 1  | schematic of the model, parametrisation and fit at each step                                                                                  |
| Supplementary 2  | unvaccinated cohorts at 0.5, 1, 2 and 10 infections per year                                                                                  |
| Supplementary 3A | validation using RTS,S with and without 1 booster                                                                                             |
| Supplementary 3B | validation using R21 with and without SMC (datoo)                                                                                             |
| Supplementary 4  | R21 for 2 years and RTS,S for 2 years                                                                                                         |
| Figure 1         | R21 for 2 years, R21+SMC for 2 years and R21+SMC for 5 years                                                                                  |
| Figure 2         | RH5 for 2 years, RH5 with x 2 half-life for 2 years and RH5 with x 2 half-life for 5 years                                                    |
| Supplementary 5  | as figure 2 but RH5 protects against blood infection                                                                                          |
| Figure 3         | R21, RH5 and R21+RH5 for 2 years                                                                                                              |
| Supplementary 6  | as figure 3 but for 5 years                                                                                                                   |
| Figure 4         | lifetime episodes averted and efficacies across transmission intensities. R21, RH5 and R21+RH5 for 2 years                                    |
| Supplementary 7  | as figure 4 but for 5 years                                                                                                                   |
| Supplementary 8  | as figure S7 but for perennial transmission                                                                                                   |
| Figure 5         | R21 rescue, R21+SMC for 5 years or R21+SMC for 5 years followed by RH5 with no boosters or R21+SMC for 5 years followed by RH5 with 1 booster |

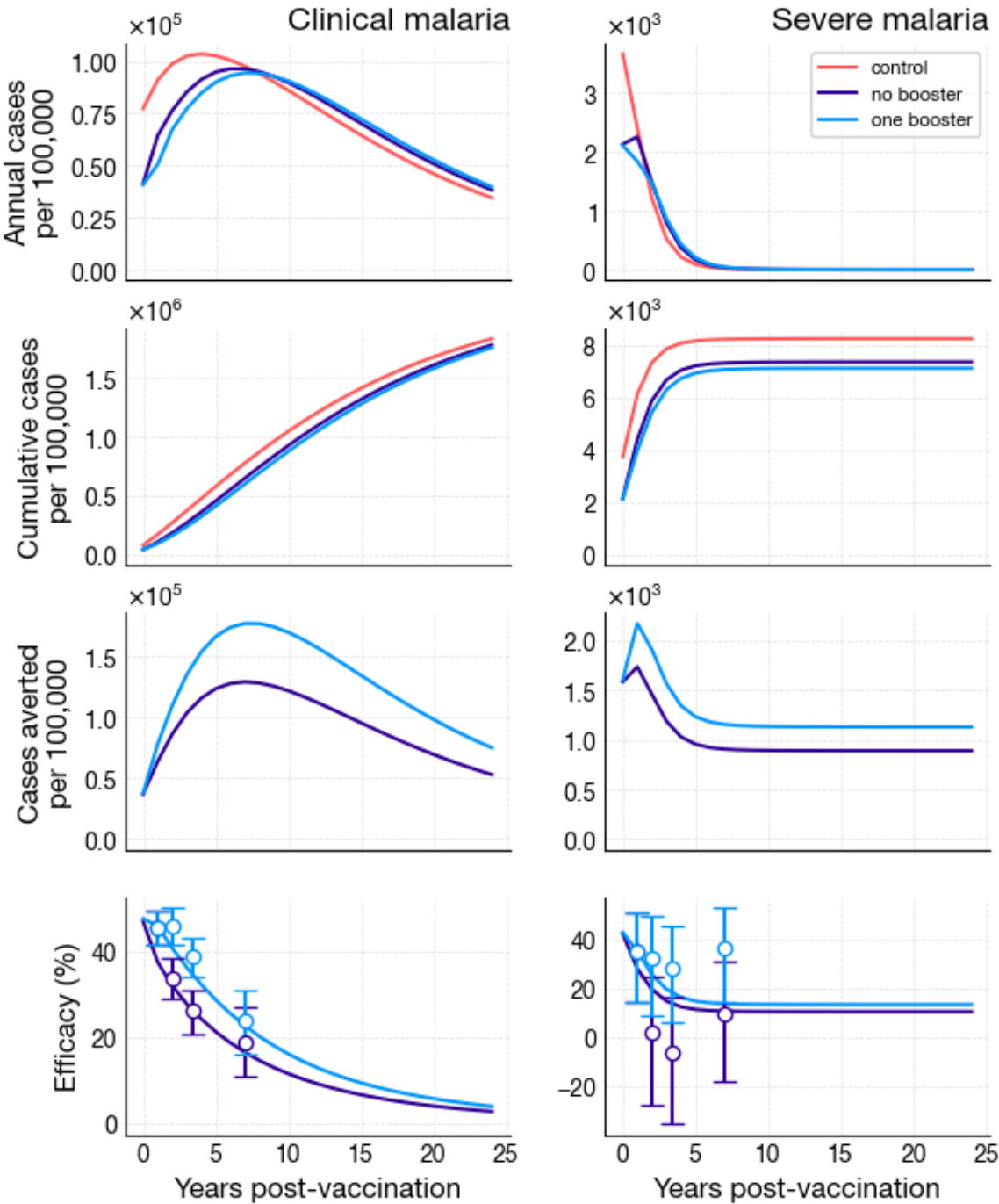
Supplementary figure 2

- Unvaccinated cohorts



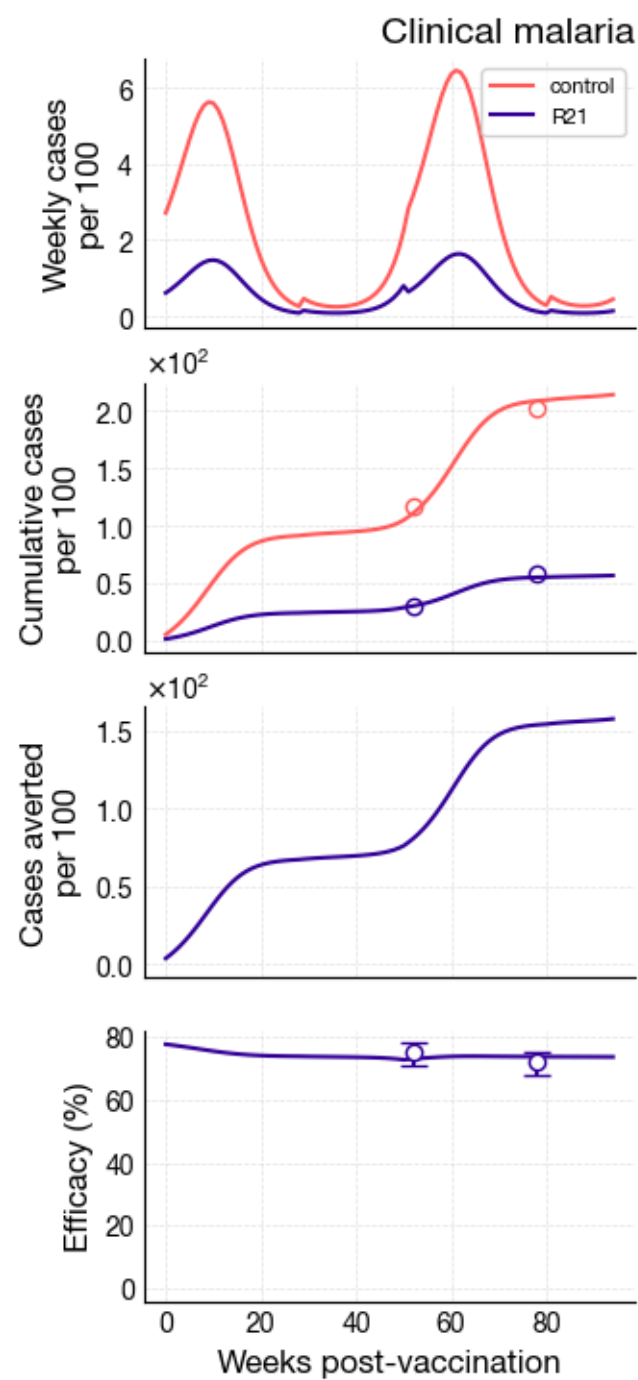
Supplementary figure 3A

• RTS,S validation



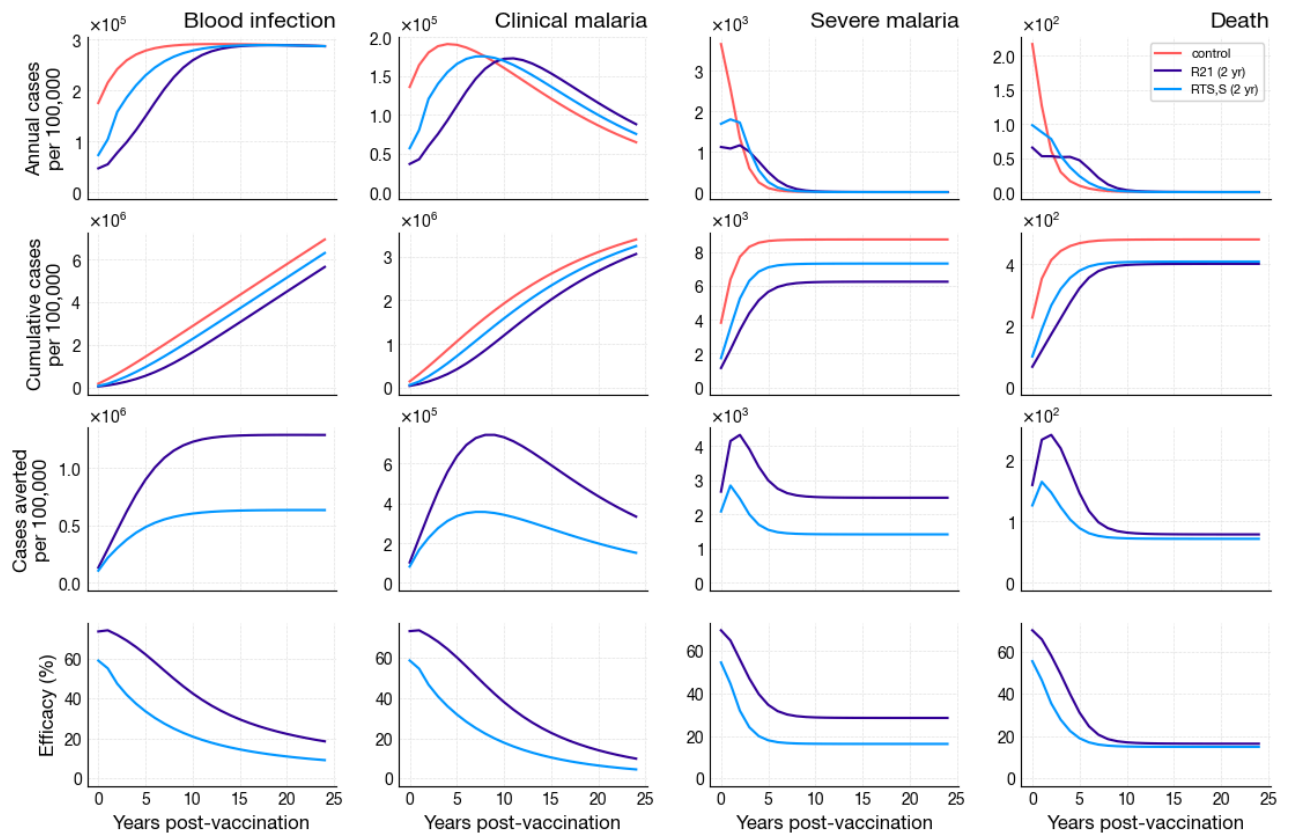
# Supplementary figure 3B

- R21 validation



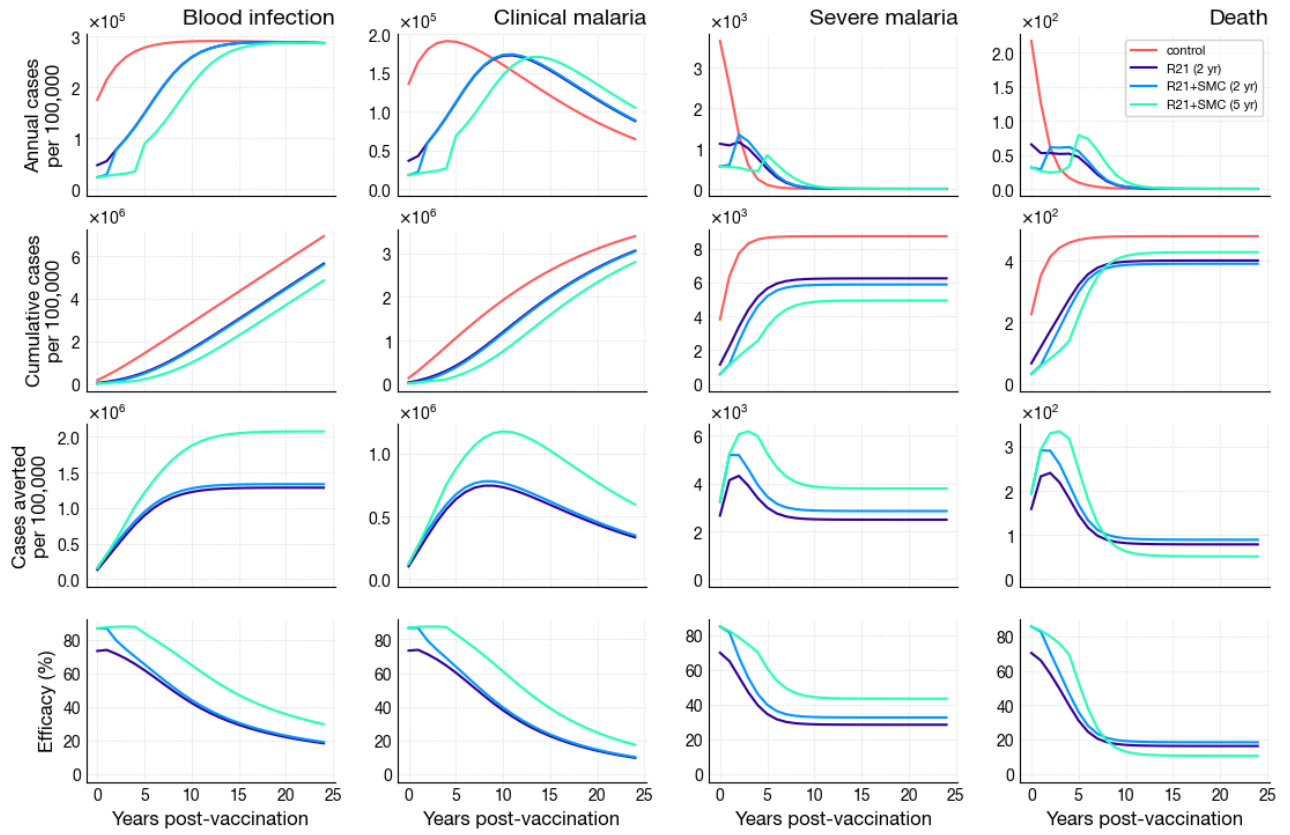
# Supplementary figure 4

- R21 for 2 years
- RTS,S for 2 years



**Figure 1**

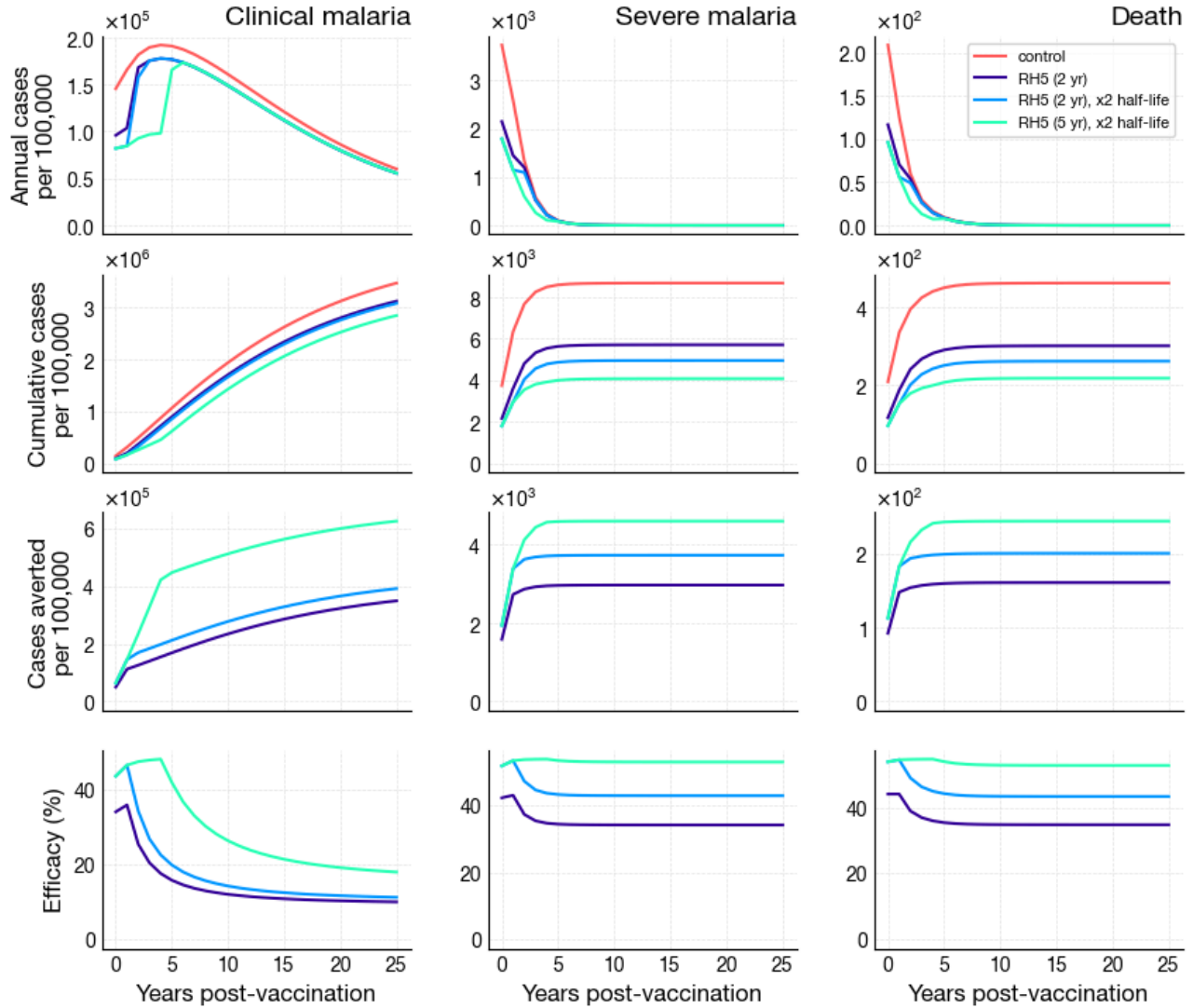
- R21 for two years
- R21+SMC for two years
- R21+SMC for five years



## Figure 2:

RH5.1 does not reduce the risk of blood infection

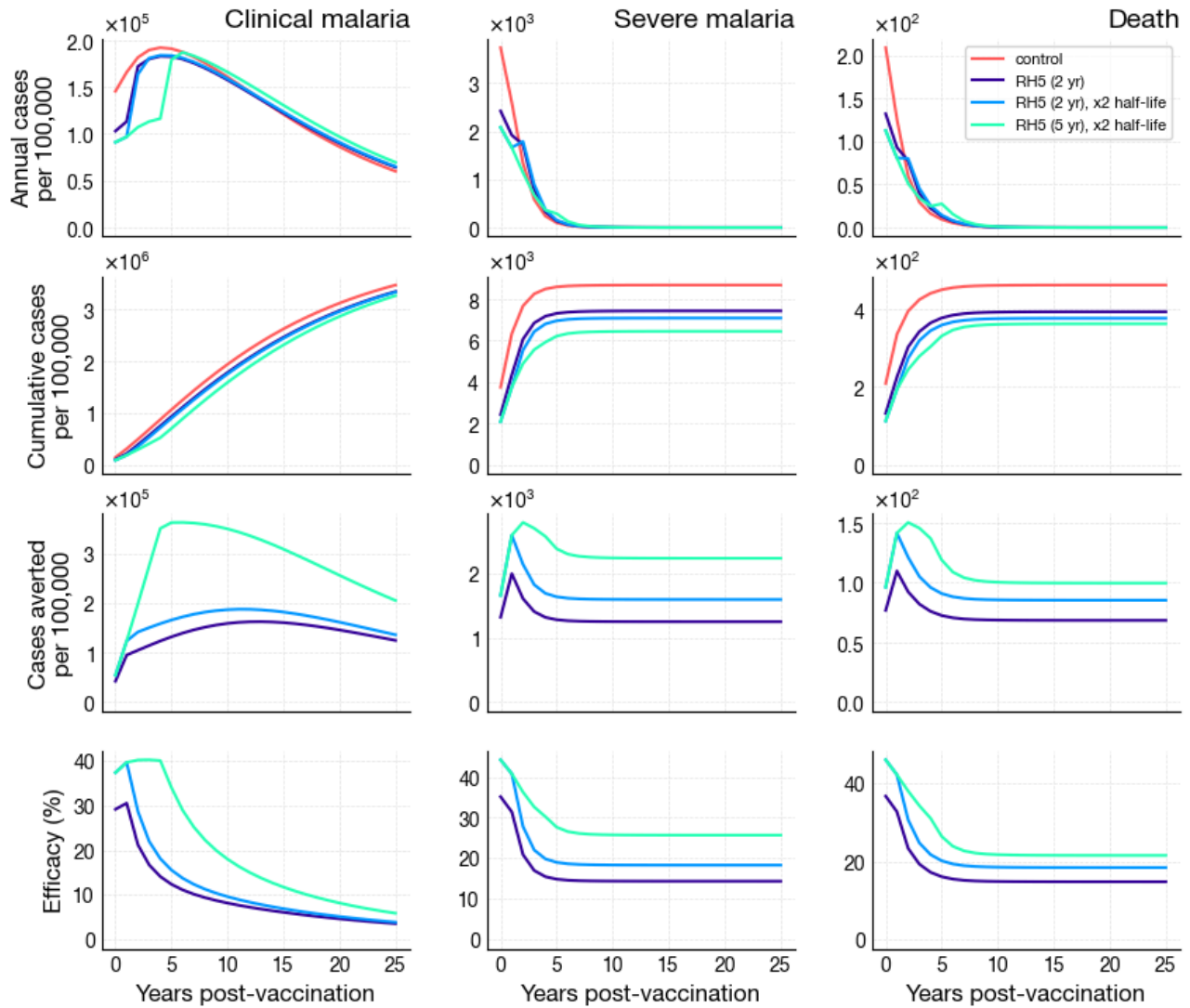
- RH5.1 for two years
- RH5.1 with x 2 half-life for two years
- RH5.1 with x 2 half-life for five years



## Supplementary figure 5:

RH5 reduces the risk of blood infection

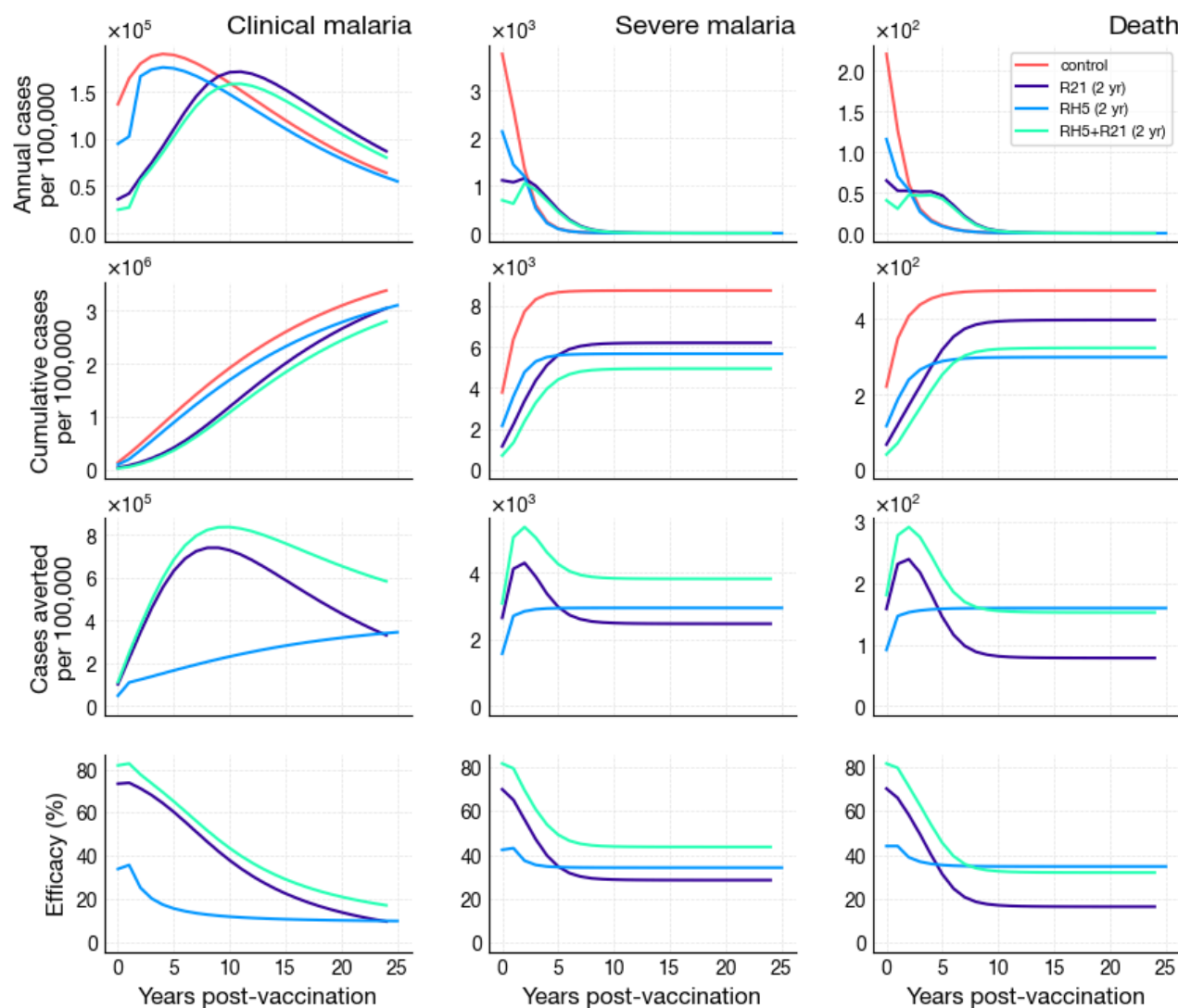
- RH5 for two years
- RH5 with x 2 half-life for two years
- RH5 with x 2 half-life for five years





**Figure 3**

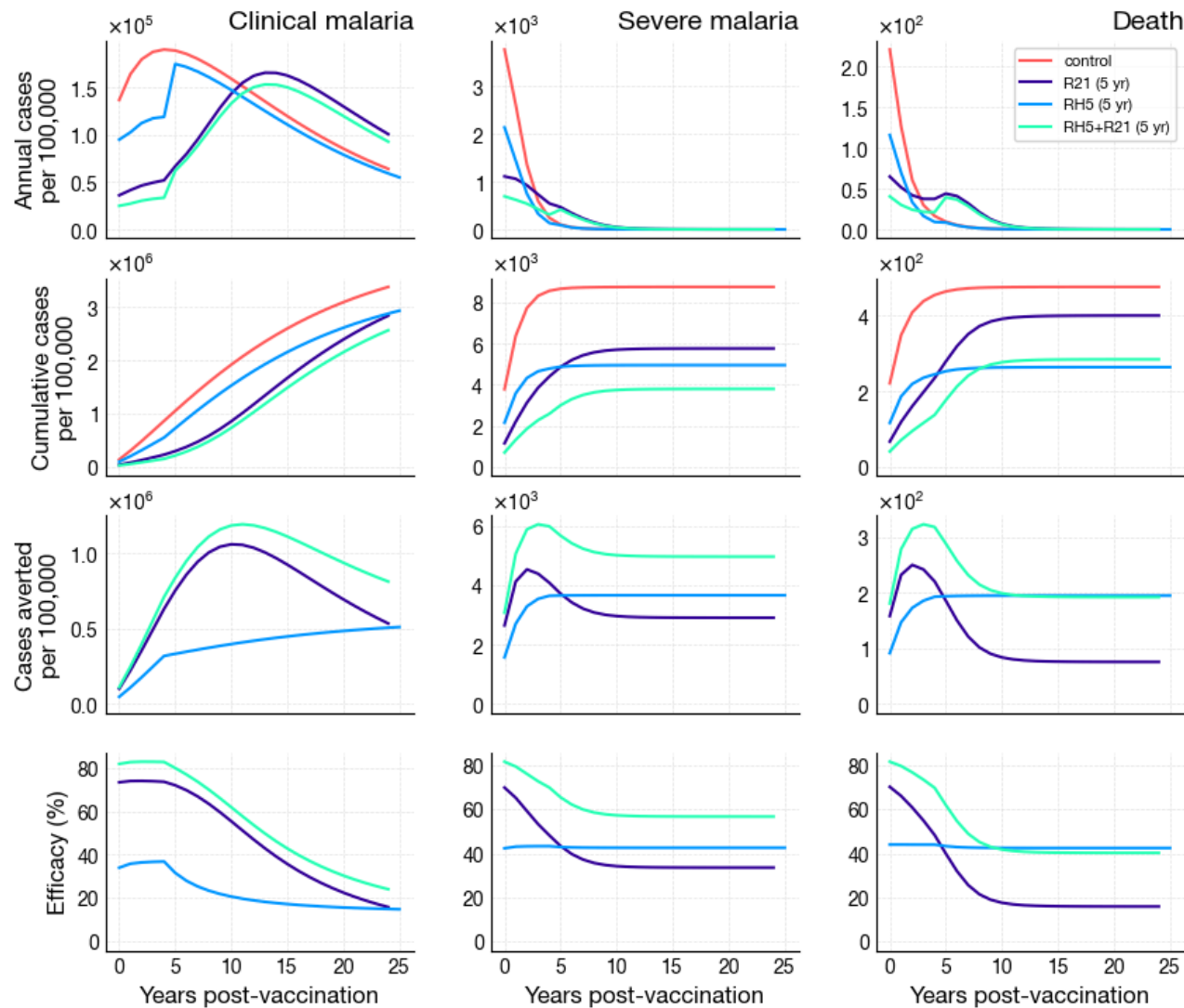
- R21 for two years
- RH5.1 for two years
- RH5.1+R21 simultaneously for two years



# Supplementary figure 6

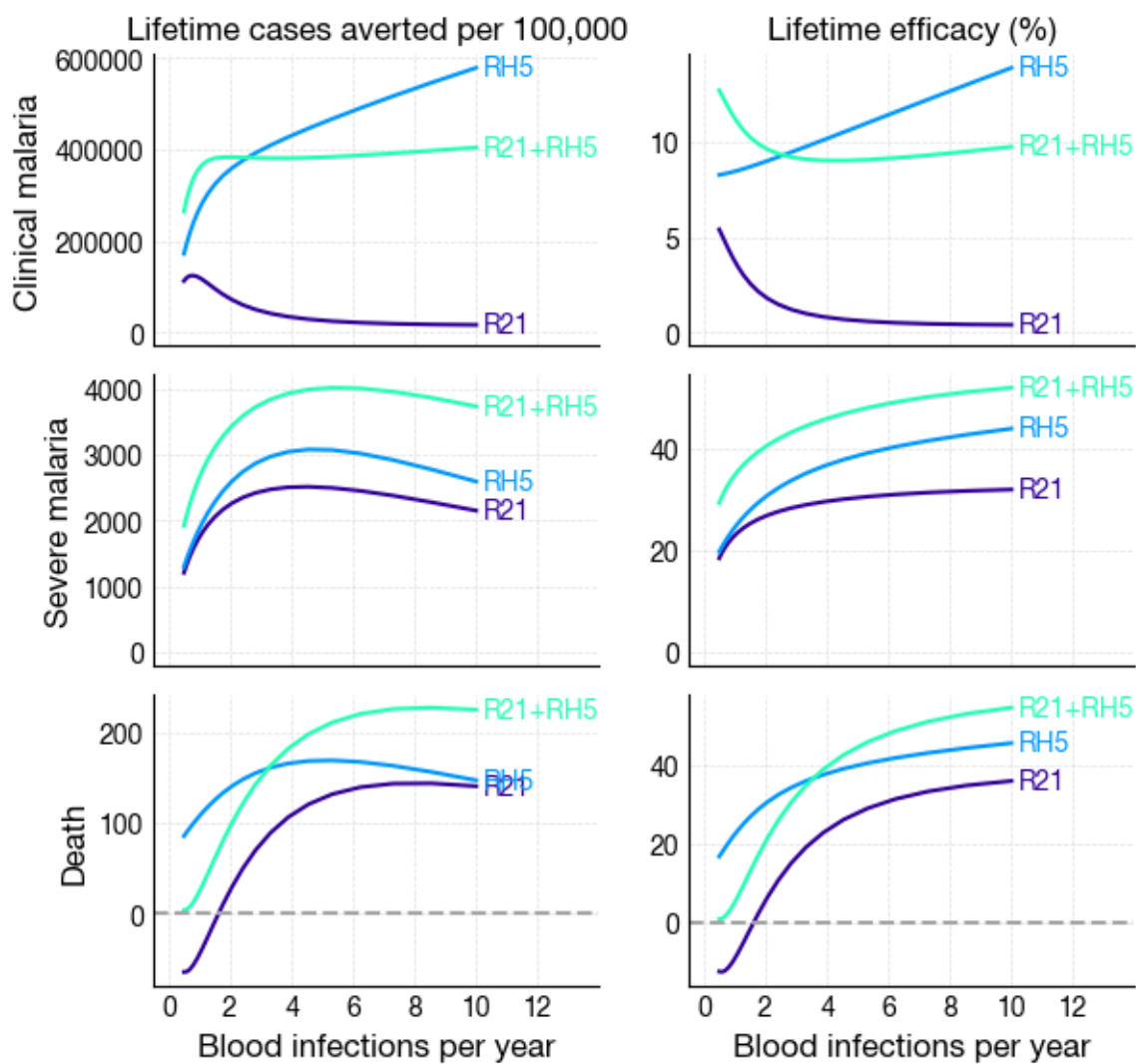
As Figure 3 but for five years

- R21 for five years
- RH5 for five years
- RH5+R21 simultaneously for five years



**Figure 4**

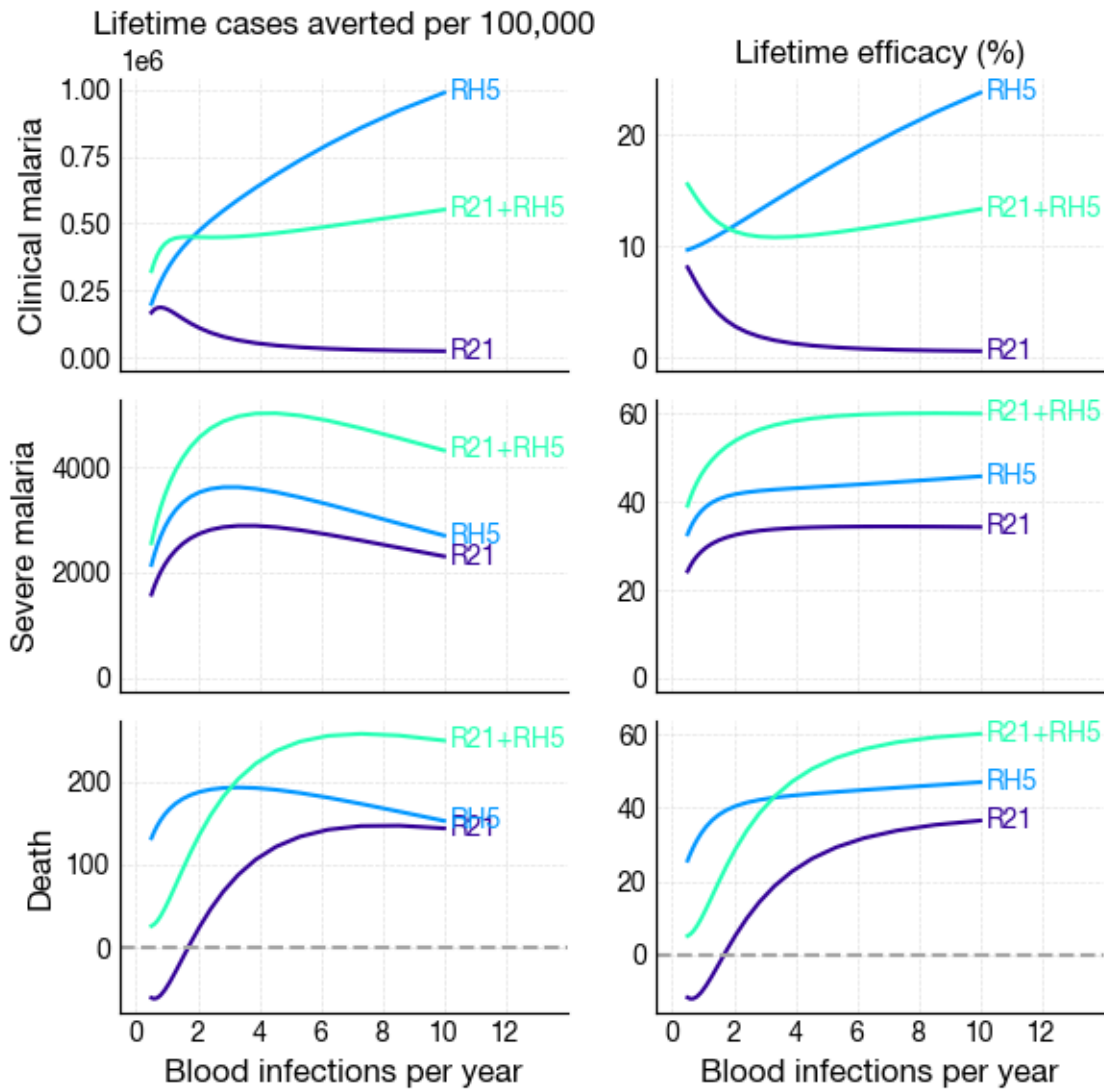
- R21, two years
- RH5.1, two years
- R21+RH5.1 simultaneously, two years



# Supplementary figure 7

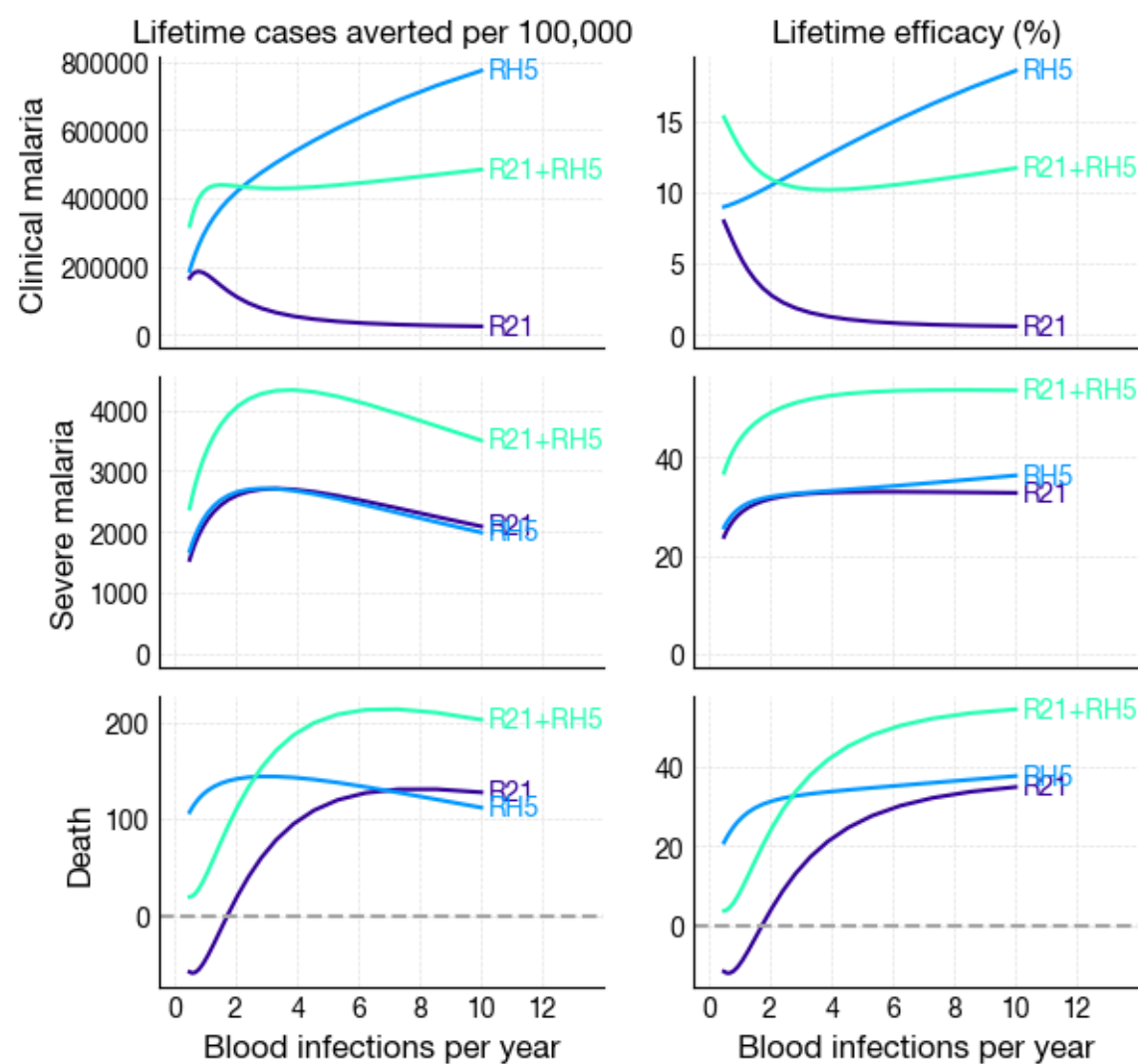
As Fig. 4 but vaccination for five years

- R21, five years
- RH5.1, five years
- R21+RH5.1 simultaneously, five years



Supplementary figure 8

As Fig. S7 but perennial transmission.



## Figure 5

Rescuing R21 with RH5.1.

- R21 alone for two years
- R21 + SMC for two years
- R21 + SMC for five years

